Table S8 Chemical-diversity and activity-threshold sensitivity

A. CAP5 membership

Target pair	Original n	CAP5 n	Excluded n	Excluded dual	Excluded A-only	Excluded B-only	Excluded neither
EGFR/HER2	109	109	0	0	0	0	0
JAK1/JAK2	110	107	3	0	3	0	0
JAK1/TYK2	110	110	0	0	0	0	0
PIK3CA/mTOR	110	110	0	0	0	0	0
AChE/BChE	98	98	0	0	0	0	0
F2/F10	110	110	0	0	0	0	0
PPARG/PPARA	110	110	0	0	0	0	0
PPARA/PPARD	110	107	3	0	1	2	0

CAP5 is a post hoc sensitivity analysis: at most five ligands per nonempty Murcko scaffold within pair and activity class; acyclic ligands are exempt. Six ligands are excluded without replacement. Complete CAP5 method comparisons, chemical models, controls and ranking outputs are retained in Supplementary_Data.zip. The PPARA/PPARD S5 dual-versus-A-only CI changes from [0.4921875, 0.767578125] to [0.5030241935, 0.7772177419]; this boundary change is retained, without a general significance claim. The strict Q110 subset has 56 members (25 dual, 12 A-only, 14 B-only, 5 neither). Main and subset populations differ; no new paired CI for their difference is calculated.

Table S8 Chemical-diversity and activity-threshold sensitivity

B. Vina minimum AUROC before and after CAP5

Target pair	Primary minimum AUROC (95% CI)	CAP5 minimum AUROC (95% CI)	Point change
EGFR/HER2	0.2467 [0.1328, 0.3783]	0.2467 [0.1328, 0.3783]	0.0000
JAK1/JAK2	0.5908 [0.4512, 0.7246]	0.6315 [0.4871, 0.7597]	0.0406
JAK1/TYK2	0.4219 [0.2812, 0.5635]	0.4219 [0.2812, 0.5635]	0.0000
PIK3CA/mTOR	0.5801 [0.4346, 0.7129]	0.5801 [0.4346, 0.7129]	0.0000
AChE/BChE	0.6341 [0.4607, 0.7540]	0.6341 [0.4607, 0.7540]	0.0000
F2/F10	0.3623 [0.2305, 0.4805]	0.3623 [0.2305, 0.4805]	0.0000
PPARG/PPARA	0.6021 [0.4561, 0.7051]	0.6021 [0.4561, 0.7051]	0.0000
PPARA/PPARD	0.5107 [0.3652, 0.6475]	0.4969 [0.3427, 0.6354]	-0.0139

CAP5 is a post hoc sensitivity analysis: at most five ligands per nonempty Murcko scaffold within pair and activity class; acyclic ligands are exempt. Six ligands are excluded without replacement. Complete CAP5 method comparisons, chemical models, controls and ranking outputs are retained in Supplementary_Data.zip. The PPARA/PPARD S5 dual-versus-A-only CI changes from [0.4921875, 0.767578125] to [0.5030241935, 0.7772177419]; this boundary change is retained, without a general significance claim. The strict Q110 subset has 56 members (25 dual, 12 A-only, 14 B-only, 5 neither). Main and subset populations differ; no new paired CI for their difference is calculated.

Table S8 Chemical-diversity and activity-threshold sensitivity

C. PIK3CA/mTOR strict 6.5/5.5 subset

Target pair	Scheme	n (dual / A-only / B-only)	AUROC, target B (95% CI)	AUROC, target A (95% CI)	Minimum AUROC (95% CI)
PIK3CA/mTOR	S1	25 / 12 / 14	0.7900 [0.6200, 0.9300]	0.4571 [0.2743, 0.6429]	0.4571 [0.2743, 0.6429]
PIK3CA/mTOR	S2	25 / 12 / 14	0.6200 [0.4133, 0.8133]	0.5343 [0.3429, 0.7229]	0.5343 [0.3343, 0.6829]
PIK3CA/mTOR	S3	25 / 12 / 14	0.7500 [0.5767, 0.8933]	0.6286 [0.4457, 0.7971]	0.6286 [0.4429, 0.7800]
PIK3CA/mTOR	S4	25 / 12 / 14	0.6267 [0.4267, 0.8167]	0.6057 [0.4143, 0.7829]	0.6057 [0.3867, 0.7229]
PIK3CA/mTOR	S5	25 / 12 / 14	0.5933 [0.4033, 0.7833]	0.6686 [0.4828, 0.8371]	0.5933 [0.4000, 0.7367]

CAP5 is a post hoc sensitivity analysis: at most five ligands per nonempty Murcko scaffold within pair and activity class; acyclic ligands are exempt. Six ligands are excluded without replacement. Complete CAP5 method comparisons, chemical models, controls and ranking outputs are retained in Supplementary_Data.zip. The PPARA/PPARD S5 dual-versus-A-only CI changes from [0.4921875, 0.767578125] to [0.5030241935, 0.7772177419]; this boundary change is retained, without a general significance claim. The strict Q110 subset has 56 members (25 dual, 12 A-only, 14 B-only, 5 neither). Main and subset populations differ; no new paired CI for their difference is calculated.

Table S8 Chemical-diversity and activity-threshold sensitivity

D. Chemistry-model sensitivity to CAP5

Target pair	Direction	Original OOF n	CAP5 OOF n	Original ECFP4 AUROC	CAP5 ECFP4 AUROC	ECFP4 AUROC change	Original Vina increment	CAP5 Vina increment
EGFR/HER2	Dual vs A-only	65	65	0.8002	0.8002	0.0000	-0.0164	-0.0164
EGFR/HER2	Dual vs B-only	59	59	0.8790	0.8790	0.0000	-0.0023	-0.0023
JAK1/JAK2	Dual vs A-only	64	61	0.9185	0.9079	-0.0106	0.0015	0.0048
JAK1/JAK2	Dual vs B-only	64	64	0.9678	0.9678	0.0000	0.0029	0.0029
JAK1/TYK2	Dual vs A-only	63	63	0.8488	0.8488	0.0000	-0.0050	-0.0050
JAK1/TYK2	Dual vs B-only	63	63	0.9042	0.9042	0.0000	-0.0010	-0.0010
PIK3CA/mTOR	Dual vs A-only	64	64	0.8213	0.8213	0.0000	0.0215	0.0215
PIK3CA/mTOR	Dual vs B-only	64	64	0.7285	0.7285	0.0000	-0.0029	-0.0029
AChE/BChE	Dual vs A-only	52	52	0.8993	0.8993	0.0000	-0.0044	-0.0044
AChE/BChE	Dual vs B-only	55	55	0.8201	0.8201	0.0000	-0.0053	-0.0053
F2/F10	Dual vs A-only	63	63	0.9405	0.9405	0.0000	-0.0071	-0.0071
F2/F10	Dual vs B-only	63	63	0.6633	0.6633	0.0000	0.0363	0.0363
PPARG/PPARA	Dual vs A-only	63	63	0.8206	0.8206	0.0000	-0.0141	-0.0141
PPARG/PPARA	Dual vs B-only	64	64	0.6670	0.6670	0.0000	-0.0059	-0.0059
PPARA/PPARD	Dual vs A-only	64	63	0.9316	0.9345	0.0028	-0.0029	-0.0030
PPARA/PPARD	Dual vs B-only	64	62	0.8740	0.8688	-0.0053	0.0049	0.0042

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